

Design of Thermal Earring: A Low-Power Wireless Earring for Longitudinal Temperature Monitoring

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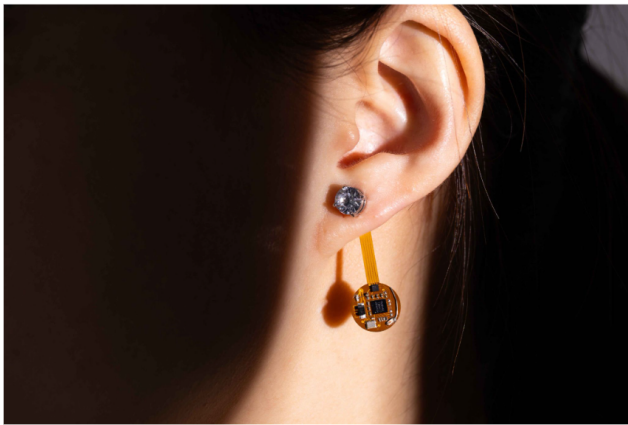
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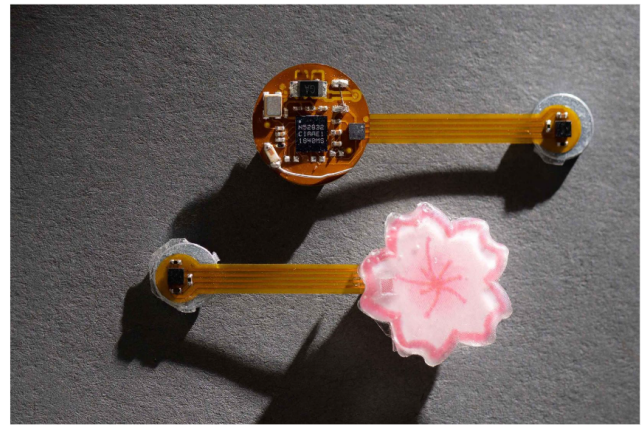
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(a)



(b)

Figure 1: (a) Thermal Earring worn by a person. (b) Thermal Earring showcased with a cherry blossom fashion design.

ABSTRACT

We present Thermal Earring, a pioneering smart earring that offers a practical and comfortable solution for wearable health monitoring. Unlike watches and other wearables typically worn loosely on body extremities, earring's proximity to the head enables unique sensing advantages. This research aims to provide design guidelines for developing smart earring and jewelry. Our Thermal Earring prototype is designed to resemble traditional earrings in size (11.3 mm in width and 31 mm in length) and weight (335 mg). Thermal Earring consumes only 14.4 uW and enables a battery life of 28 days, making it suitable for real world use. We demonstrate that its

compact and lightweight design allows it to seamlessly integrate into fashionable jewelry.

CCS CONCEPTS

• Human-centered computing → Ubiquitous and mobile computing systems and tools; • Applied computing → Consumer health.

KEYWORDS

Smart Jewelry; Smart Earring; Wearable Computing; Health Monitoring

ACM Reference Format:

Qiuyue (Shirley) Xue, Yujia Liu, Vikram Iyer, and Shwetak Patel. 2024. Design of Thermal Earring: A Low-Power Wireless Earring for Longitudinal Temperature Monitoring. In *Companion of the 2024 ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp Companion '24)*, October 5–9, 2024, Melbourne, VIC, Australia. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3675094.3681953>

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UbiComp Companion '24, October 5–9, 2024, Melbourne, VIC, Australia

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ACM ISBN 979-8-4007-1058-2/24/10

<https://doi.org/10.1145/3675094.3681953>

1 INTRODUCTION

In recent years, wearable devices have become increasingly popular and ubiquitous in daily life. These devices, which remain in close contact with the user's body throughout the day, offer unique opportunities for health monitoring, interaction, and activity tracking. This has inspired the exploration of various accessories, including smart rings [3, 16], smart glasses [26], fitness earbuds [7, 18], and smart clothing [21, 24]. However, we note that a significant category of everyday accessories, worn by millions, has been largely overlooked in the research community: jewelry. For instance, in the United States, 76% of women have pierced earlobes—a rate nearly double that of smartwatch adoption. Additionally, the growing trend of men wearing earrings highlights the widespread acceptance of earrings as fashionable accessories.

Earrings, in particular, present a unique sensing opportunity for physiological signals given the tight attachment and unique placement. Comparing to headphones or earbuds, earrings are often worn throughout most of the day, making them ideal for continuously monitoring physiological signals. Moreover, comparing to watches, earrings are more tightly attached to the body which offers special sensing advantages. For example, emotions such as embarrassment or stress can cause increased blood flow to the face and ears, resulting in flushed cheeks and temperature changes. Monitoring body temperature continuously is crucial, yet challenging, as traditional methods like oral, axillary, in-ear, or skin thermometers only provide intermittent readings. These sporadic measurements may miss critical temperature fluctuations, such as those caused by a fever during a viral infection. Continuously tracking fine-grained temperature changes could provide new insights into daily activities and health signals.

While developing a smart earring with continuous physiological sensing on earlobe provides many sensing benefits, it presents multiple technical challenges. The earring system must be compact and lightweight for comfortable use. These constraints on size inherently limits the power consumption of the system due to the restricted energy density of batteries. Taking these strict constraints into design consideration, we introduce Thermal Earring, a first-of-its-kind wireless smart earring that enables continuous temperature monitoring. Thermal Earring measures a maximum width of 11.3 mm and a length of 31 mm, weighs 335 mg, and with a battery life of a month. By monitoring the earlobe's temperature continuously, Thermal Earring enables unique applications such as fever monitoring, stress and emotion detection, and monitoring daily activities like eating and exercising.

2 RELATED WORK

Smart jewelry is an emerging class of wearable devices that seeks to combine fashion and function by integrating sensing into jewelry accessories. Vega et al. have highlighted the concept of integrating technology with beauty products, proposing wearable computing in forms like artificial nails and makeup [25]. Previous research in smart jewelry has explored various functional items: smart earrings [17, 19], necklaces [5, 12, 27, 28], bracelets [2, 8, 9, 22], glasses [1, 6, 15, 23], rings [3, 14, 16, 20, 29, 30], nails [10, 13], and nose rings [4]. However, previous smart jewelry devices often feature larger than

desired form factors. Designing smart jewelry that preserves the compact size of traditional jewelry remains a significant challenge.

3 GENERAL DESIGN PRINCIPLES

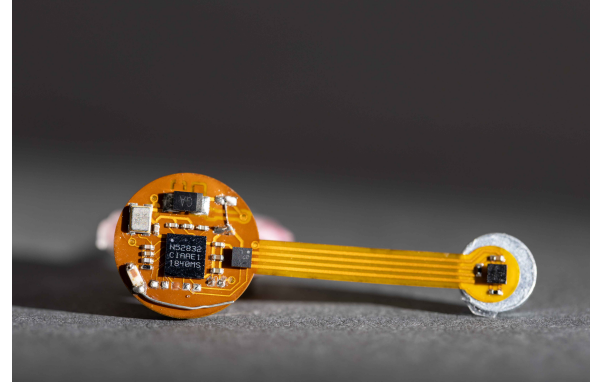


Figure 2: The Thermal Earring system, an example of the smart earring design. The Thermal Earring prototype shown here is about the size and weight of a small paperclip and has a 28-day battery life.

We summarize the lessons learned during the development of the Thermal Earring and examine the key considerations for designing smart earrings and other tiny wearable devices.

3.1 Smart Earring Design Considerations

Creating compact, low-power wearables like smart jewelry brings unique advantages and significant challenges. Key considerations include managing size, weight, and power consumption, all of which are often tightly interrelated and critical for ensuring comfort.

- **Compact Size:** The earring must be small enough to fit comfortably on the earlobe.
- **Lightweight:** The earring should be light to avoid any discomfort during long time wear.
- **Reasonable Battery Life:** Despite form factor constraints, the earring should offer a long battery life without frequent recharging.
- **Flexibility:** Similar to common dangling earrings, the smart earring should accommodate flexibility for the dangling part.

3.2 Minimizing Size and Weight

Smart jewelry, especially earrings, must be compact and lightweight to ensure comfort and wearability. Since most components have similar densities, their size directly affects their weight. These constraints limit the available space for essential components. Therefore, designing such devices requires careful selection and integration of the smallest and lightest components to maintain both functionality and comfort.

3.2.1 Compact Components. It is important choose sensors and microcontrollers available in compact wafer packages, some of which include built-in wireless features like Bluetooth and WiFi. However, smaller chips typically have limitations in memory and processing power. Careful consideration is needed to balance these trade-offs while maintaining the device's performance and functionality.

3.2.2 Efficient PCB Design. Integrating multiple functionalities (e.g., sensors, wireless communication, and processing) into a tiny form factor requires highly optimized PCB layouts to maximize space efficiency. Optimizing the PCB design is crucial for fitting all necessary functionalities into a small form factor.

3.3 Power Management

We summarize the techniques we have explored to minimize power consumption without compromising performance.

3.3.1 Balance On-device Computing. Offload complex tasks to paired smartphones or computers, leaving the microcontroller to handle only simple tasks like sensor interfacing and wireless transmission.

3.3.2 Optimize Wireless Efficiency. Wireless transmissions are typically the most energy-intensive aspect of wearables, drawing high current (in mA) during short transmission periods (lasting milliseconds). Transmission time and power consumption are directly related to the length of the transmitted data. Therefore, it can be beneficial to process some data on the wearable itself to reduce its size before transmission. For example, calculating a single heart rate value from sensor data before sending it minimizes data size and lowers power consumption, despite requiring some additional processing on the device.

3.3.3 Duty Cycle and Deep Sleep. Effective power management strategies also include putting the microcontroller into deep sleep mode when it's not actively interfacing with sensors or transmitting data. In practice, this method can significantly reduce power consumption, potentially by a factor of hundreds, depending on the wake-up frequency.

3.3.4 Power harvesting. Power harvesting can supplement batteries by leveraging ambient energy sources like solar, kinetic, and RF energy. Since harvested energy is often variable and intermittent, combining it with traditional batteries as the primary source provides both a stable power supply and a sustainable solution.

3.4 Integration with Fashion

Smart jewelry must be compact and lightweight to blend seamlessly with various fashion designs. The technology can be encased in electrical isolation and waterproof layers and then embedded into molds made of common materials like metal, resin, and ceramics. Techniques such as welding or attaching precious metals and gemstones can enhance its aesthetic appeal. For charging, smart jewelry can be placed in a jewelry box with magnetic attachments, similar to AirPods, or use wireless charging methods.

By experimenting with different shapes, materials, and colors, smart jewelry can be customized to match the diverse styles of traditional jewelry, ensuring it remains both practical and fashionable.

4 THERMAL EARRING DESIGN

Thermal Earring achieves the design considerations discussed above and present a practical wearable platform for longitudinal health sensing. Inspiration from traditional dangle earrings, Thermal Earring's temperature sensing unit is placed on the earlobe, while larger components such as the microcontroller and battery, are placed on the dangling part. As shown in Figure 2, this design

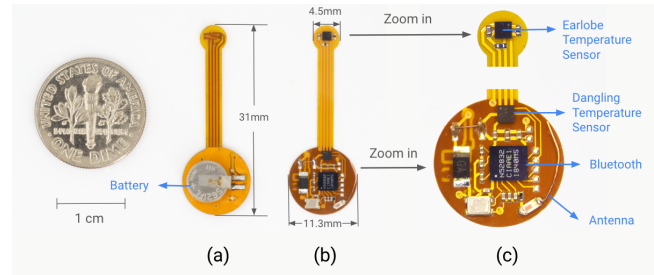


Figure 3: The Thermal Earring system with one dime coin as reference. (a) The back side of the Thermal Earring. (b) the front side of the Thermal Earring. (c) The zoomed-in view of the Thermal Earring components.

achieves an appearance of a typical dangle earring while providing a functional platform. Below, we will detail the key components of the Thermal Earring system, including the temperature sensors, microcontroller, and battery.

4.1 Dual Temperature Sensors Design

The Thermal Earring incorporates a dual temperature sensor design that enables concurrent measurement of both earlobe and ambient temperatures. By distinguishing between variations in the user's core body temperature and those in the surrounding environment, this design enhances the accuracy of physiological monitoring. HDC2010 temperature sensor from Texas Instruments is chosen for its small size (1.5 mm x 1.5 mm) and low power consumption (0.9 uW). One sensor directly contacts the earlobe to measure skin temperature, while the other is placed in the dangling part for sensing environmental temperature. This configuration effectively mitigates the influence of ambient temperature fluctuations on sensing body temperature, for example, transitioning from indoor to outdoor settings results in rapid changes of the temperature reading. This precision is crucial for applications like ovulation tracking which requires reliable and continuous temperature monitoring.

4.2 Microcontroller and Wireless Communication

Wirelessly transmitting the temperature data collected by the wearable to a smartphone or computing device for further analysis is needed, but is also the most power hungry part of a wearable system. After carefully considering the size and power consumption of different microcontrollers, we chose nRF52832 microcontroller chip for computing and wireless communication. This ultra-compact chip comes in a 3 mm x 3 mm package and consumes 5.7 uW in system on mode.

The two temperature sensors are connected to nRF52832 through I2C with different addresses for synchronized data acquisition. The temperature data collected is packed into advertising packets and transmitted to a smartphone. We utilize the advertising mode of nRF52832 Bluetooth chip since it is more energy-efficient than setting up constant Bluetooth connection with the smartphone. The Bluetooth chip is configured to enter sleep mode once after interfacing with the sensors and transmitting Bluetooth to reserve power consumption.

4.3 Power Source and Battery Life

Batteries are typically the largest and heaviest components in small wearable devices due to the energy constraints of current battery technology. We chose the Seiko MS621FE lithium rechargeable battery to meet our form factor requirements. This battery has a diameter of 6.8 mm and weighs only 0.23 grams. However, the small size of the battery limits its capacity to 5.5 mAh. By carefully optimizing power efficiency, Thermal Earring achieves an ultra-low average current of 4.8 μ A, resulting in a battery life of 28 days.

Despite its compact size, the battery's discharge current is limited to 15 μ A, enough for sleep mode but insufficient for Bluetooth transmissions, which require at least 5 mA. To address this, we added a 100 μ F low ESR capacitor (F980G107MSA) in parallel with the battery. This capacitor charges while the system is in deep sleep and discharges quickly during Bluetooth transmissions, supplying the necessary current bursts for mA-level pulses.

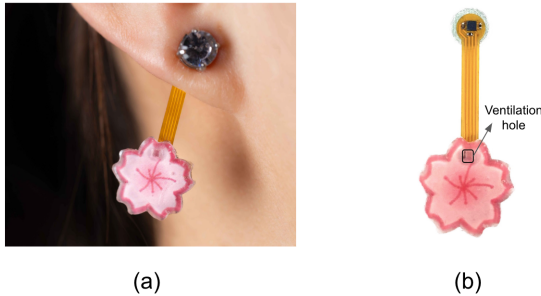


Figure 4: (a) A person wearing Thermal Earring with cherry blossom design. (b) Thermal Earring with cherry blossom design.

4.4 The Final Earring System

The final prototype of the Thermal Earring, shown in Figure 3, features a flexible printed circuit board (PCB) that mimics the flexibility of traditional earrings. The ambient temperature sensor and Bluetooth microcontroller are on the front of the dangling part, and the battery is on the back. The earlobe temperature sensor is on the segment attached to the earlobe. The earring is secured to user's earlobe using commercial magnetic earrings. A Kapton layer insulates the electronics from the skin. The earring system is compact: 4.5 mm in diameter for the earlobe part, 11.3 mm for the dangling part, 31 mm in total length, and 3.46 mm thick. Weighing only 0.335 grams, it is much lighter than the average commercial earring (around 3 grams)[11]. This design allows for future customization, including artistic enclosures and gemstone integration.

4.5 Fashion Design

We created a cherry blossom earring using resin, demonstrating how the Thermal Earring can blend style and function (Figure 4). Thermal Earring's flat, thin, and flexible features allows it to be embedded in resin molds of different shapes and colors. For this example, we used a cherry blossom-shaped silicone mold, embedding the Thermal Earring with a ventilation hole to ensure accurate ambient temperature sensing. In addition to resin, there are many

other methods can be further explored for incorporating fashion design into Thermal Earring.

5 DISCUSSION

5.1 Applications and Use Cases

Through longitudinal earlobe temperature monitoring, Thermal Earring can track fevers, detect acute stress and emotional changes such as blushing, as well as daily activities like eating and exercising. These features provide insights into the user's overall wellness.

In addition to offering real-time health data and wellness feedback to users, Thermal Earring also serves as a continuous health sensing platform for researchers to develop future technologies for automatic activity recognition and disease detection.

5.2 Future Directions

While Thermal Earring specifically focuses on temperature monitoring, it opens up new sensing possibilities for wearable wellness from the earlobe. Future directions could expand its capabilities to include motion tracking, exercise monitoring, and heart rate measurement, making it a versatile health monitoring tool.

Besides earrings, other types of jewelry can be transformed into smart wearables too, yet they remain largely unexplored. Each type of jewelry can sense different areas of the body, offering unique benefits. For example, necklaces could track chest signals, and nose rings could provide detailed breathing signals. Exploring these possibilities will unlock new dimensions of smart wearables, allowing for more comprehensive health monitoring.

Health monitoring is one example application of smart earring and smart jewelry. These devices can also be used for other use cases, such as interactive input devices, on-body displays, and haptic feedback systems.

6 CONCLUSION

We present Thermal Earring, a novel small (11.3 mm in width, 31 mm in length), light-weight (0.335 grams), and ultra-low-power (one month battery life) wireless earring that continuously monitors a user's earlobe temperature. The Thermal Earring takes advantage of the unique position of earrings in proximity to the head to reliably sense body temperature, offering reliable temperature measurements from the earlobe. We explore the general principles of developing smart jewelry, sharing the methods and insights gained from our experience with Thermal Earring. We also demonstrate a fashion design of the Thermal Earring, showcasing that it can be both functional and fashionable as a smart jewelry. Overall, Thermal Earring presents an advancement in wearable health sensing, incorporating functionality, comfort, and aesthetic appeal to seamlessly integrate health monitoring into daily life.

7 ACKNOWLEDGMENTS

We thank Raymond Smith (University of Washington) for taking the pictures in Figure. 1(a) and (b), Figure. 2, and Figure. 4 (a). We thank the endowment funds provided by the Washington Research Foundation for supporting this research.

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